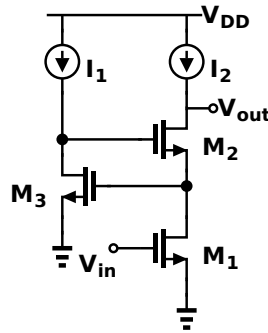
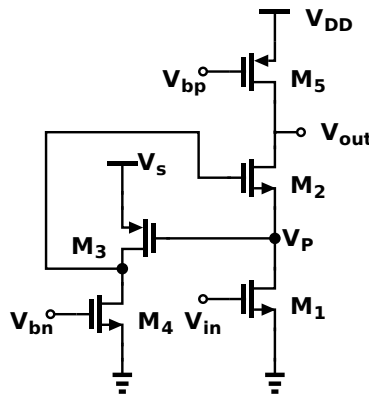


1. Suppose in the given circuit, $I_1 = 100 \mu\text{A}$, $I_2 = 0.5 \text{ mA}$, and $(W/L)_{1-3}=100/0.5$, $V_{DD}=3 \text{ V}$, $\mu_n C_{ox} = 172.35 \mu\text{A}/\text{V}^2$, $V_{thn}=0.7 \text{ V}$, $\lambda_n = 0.1 \text{ V}^{-1}$.
- Calculate the gate bias of M_2 and M_3 .
 - Calculate the overall voltage gain.
 - Now, considering the current sources I_1 and I_2 are implemented with PMOS devices having $(W/L)_p = 50/0.5$, $\mu_p C_{ox} = 51.7 \mu\text{A}/\text{V}^2$, $V_{thp}=-0.8 \text{ V}$, $\lambda_p = 0.2 \text{ V}^{-1}$ determine the output voltage swing, and the overall voltage gain.



2. Given: $V_{DD}=1.8$ V, $\mu_n C_{ox} = 150 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 100 \mu\text{A}/\text{V}^2$, $(W/L)_n=150$, $(W/L)_p=100$, $V_{thn}=0.7$ V, $V_{thp}=-0.85$ V. Assume the current flowing through M_1 is 0.1 mA.

- Estimate the value of V_{bp} .
- Suppose we are aiming for $V_P = V_{OV1}$, and $V_{out}|_{\min} = 2V_{OV1}$. With the given parameters, check the region of operation of M_3 .
- Targeting the overdrive voltage of M_4 as 0.1 V, estimate the required value of V_S .
- Detach the auxiliary amplifier of the gain boosting stage and replace M_5 with an ideal current source of 0.1 mA. Now calculate the value of λ_n required for $M_{1,2}$ to have an approximate gain of 2550 V/V. M_1 and M_2 are identical.
- Now, assume $\lambda_p = 1.3\lambda_n$, calculated from part (d). With these information, calculate the overall voltage gain of the original circuit.



3. Given Parameters: $V_{DD}=3$ V, power consumption=3 mW, $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$, $\mu_p C_{ox} = 100 \mu\text{A}/\text{V}^2$. Assume, M_1 and M_2 are identical, and have $(W/L) = \frac{100}{0.5}$. Current flowing through M_1 is $500 \mu\text{A}$. $(W/L)_5 = (W/L)_8 = 0.4 (W/L)_6$. Also, $(W/L)_3 = 0.5 (W/L)_6$; and $(W/L)_4 = (W/L)_9 = (W/L)_1$. Consider, $\lambda_n = 0.1 \text{ V}^{-1}$ and $\lambda_p = 0.2 \text{ V}^{-1}$, $V_{thn}=0.7$ V, $V_{thp}=-0.8$ V.

- If the output swing is 2.5 V, calculate $(W/L)_6$.
- Assuming V_p to be overdrive voltage of M_1 , calculate the value of resistance R_3 .
- Considering, to have a drain-to-source voltage drop across M_9 to be 15% higher than its overdrive to work as a current source, check, wheather M_4 still in saturation?
- Calculate, $(W/L)_7$, R_1 , R_2 .
- Calculate the overall gain.

